US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250283716
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Thimirachandra; Sanjeewa et al.

System and Method for Dimensioning Objects

Abstract

An example dimensioning device includes: a depth sensor configured to obtain depth data within a field of view; a processor interconnected with the depth sensor, the processor configured to: obtain depth data within the field of view including an object to be dimensioned; obtain a subset of the depth data from a predefined target region within the field of view; select a dimensioning function for the object, the dimensioning function having a target orientation of the object; validate the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, apply the dimensioning function to obtain dimensions of the object.

Inventors: **Thimirachandra; Sanjeewa (Kotugoda, LK), Tenkasi Shankar; Raghavendra (Holbrook, NY), Zhang; Ruth (Hauppauge, NY)**

Applicant: **ZEBRA TECHNOLOGIES CORPORATION** (Lincolnshire, IL)

Family ID: **96809904**

Appl. No.: **18/601597**

Filed: **March 11, 2024**

Publication Classification

Int. Cl.: **G01B21/18** (20060101); **G01B21/02** (20060101)

U.S. Cl.:

CPC **G01B21/18** (20130101); **G01B21/02** (20130101);

Background/Summary

BACKGROUND

[0001] Determining the dimensions of objects may be necessary in a wide variety of applications. For example, it may be desirable to determine the dimensions of freight, parcels, packages in a warehouse prior to shipping or storage. Different shapes and sizes of objects may be optimally dimensioned by different dimensioning functions. Small objects may be difficult to dimension as their size may cause increased distance from the dimensioning device if the user is standing while the object is on the floor, and therefore decreased resolution and accuracy in dimensioning.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0002] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

[0003] FIG. **1** is a schematic diagram of a system for dimensioning objects.

[0004] FIG. **2** is a block diagram of certain internal hardware components of the detection device of FIG. **1**.

[0005] FIG. **3** is a flowchart of a method for dimensioning objects.

[0006] FIG. **4** is a flowchart of an example method for selecting a dimensioning function and validating the target orientation at block **320** of the method of FIG. **3**.

[0007] FIG. **5**A is a schematic diagram of a performance of block **405** of the method of FIG. **4**.

[0008] FIG. **5**B is a schematic diagram of a performance of block **410** of the method of FIG. **4**.

[0009] FIG. **6** is a flowchart of an example method of applying a dimensioning function using a perpendicular-face approach at block **325** of the method of FIG. **3**.

[0010] FIGS. **7**A and **7**B are schematic diagrams of aligning an alignment indicator with a region indicator at block **620** of the method of FIG. **6**.

[0011] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0012] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0013] Examples disclosed herein are directed to a dimensioning device comprising: a depth sensor configured to obtain depth data within a field of view; a processor interconnected with the depth sensor, the processor configured to: obtain depth data within the field of view including an object to be dimensioned; obtain a subset of the depth data from a predefined target region within the field of view; select a dimensioning function for the object, the dimensioning function having a target orientation of the object; validate the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, apply the dimensioning function to obtain dimensions of the object.

[0014] Additional examples disclosed herein are directed to a method comprising: obtaining depth data from a field of view including an object to be dimensioned; obtaining a subset of the depth data from a predefined target region within the field of view; selecting a dimensioning function for the object, the dimensioning function having a target orientation of the object; validating the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, applying the dimensioning function to obtain dimensions of the object.

[0015] Additional examples disclosed herein are directed to a non-transitory computer-readable storage medium storing instructions thereon, which when executed configure a processor to: obtain depth data within a field of view including an object to be dimensioned; obtain a subset of the depth data from a predefined target region within the field of view; select a dimensioning function for the object, the dimensioning function having a target orientation of the object; validate the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, apply the dimensioning function to obtain dimensions of the object.

[0016] FIG. **1** depicts a system **100** for dimensioning objects in accordance with the teachings of this disclosure. The system **100** includes a computing device **104** (also referred to herein as the dimensioning device **104** or simply the device **104**) configured to dimension a target object **108** according to a selected dimensioning function. For example, the object **108** may be an item, such as a box or package, in a transport and logistics facility. Preferably, the target object **108** may be substantially cuboidal in shape.

[0017] The device **104** may be a mobile computing device, such as a mobile phone, a tablet, a barcode scanner, a dedicated dimensioning device, or the like. In such examples, the device **104** may include a depth sensor **112**, a set of depth sensors, or one or more additional sensors such as image sensors (e.g., optical cameras, infrared sensors, etc.), ambient light sensors, proximity sensors, temperature sensors, and the like to capture dimensioning data representing the object **108**. In other examples, the device **104** may be a fixed computing device such as a desktop computer, a kiosk, or the like, and the device **104** may be associated with the depth sensor **112** to obtain data from the sensor **112** to dimension the object **108**.

[0018] In particular the depth sensor **112** may have a field of view **116** for which the depth sensor **112** is able to capture depth data. As described herein, the device **104** may capture preliminary depth data from the field of view **116**, and then process the preliminary depth data, in particular from a predefined target region of the field of view **116**, for the dimensioning operation. The depth sensor **112** may further have a principal axis **120** substantially perpendicular to a plane of the depth sensor **112**.

[0019] The device **104** may be in communication with a server **124** via a communication link, illustrated in the present example as including wireless links. For example, the link may be provided by a wireless local area network (WLAN) deployed by one or more access points (not shown). In other examples, the server **124** is located remotely from the device **104** and the link may therefore include one or more wide-area networks such as the Internet, mobile networks, and the like. The server **124** may be any suitable server environment, including a plurality of cooperating servers operating, for example in a cloud-based environment.

[0020] The system **100** is generally deployed to dimension target objects, such as the object **108**. In particular, the dimensioning device **104** is configured to dimension substantially cuboidal target objects **108**, and further, target objects **108** which are generally sufficiently small so as to be held in the hand by a user. Small objects may typically be difficult for typical dimensioning operations which rely on reference surfaces such as the floor, a table, or other support surface, since dimensioning device may not have sufficient resolution at a normal operating distance to accurately assess the dimensions of the small object. Accordingly, as described herein, the dimensioning device **104** is configured to allow a user to hold the target object **108** and performs a dimensioning operation on a small object. For example, the dimensioning device **104** is configured to obtain preliminary depth data from the field of view **116** of the depth sensor **112**, where the target object **108** is in the field of view **116**. The dimensioning device **104** may select a dimensioning function and validate a target orientation of the object **108** for the selected dimensioning function using a subset of the preliminary depth data corresponding to a predefined target region of the field of view **116**. The dimensioning device **104** may facilitate alignment in the target orientation when the target orientation is perpendicular to the principal axis **120** by displaying an alignment indicator on the device **104**.

[0021] Turning now to FIG. **2**, certain internal components of the dimensioning device **104** are illustrated. The device **104** includes a processor **200** interconnected with a non-transitory computer-readable storage medium, such as a memory **204**. The memory **204** includes a combination of volatile memory (e.g. Random Access Memory or RAM) and non-volatile memory (e.g. read only memory or ROM, Electrically Erasable Programmable Read Only Memory or EEPROM, flash memory). The processor **200** and the memory **204** may each comprise one or more integrated circuits.

[0022] The memory **204** stores computer-readable instructions for execution by the processor **200**. In particular, the memory **204** stores an application **208** which, when executed by the processor, configures the processor **200** to perform various functions discussed below in greater detail and related to the dimensioning operation of the device **104**. Some or all of the application **208** may also be implemented as a suite of distinct applications.

[0023] Those skilled in the art will appreciate that the functionality implemented by the processor **200** may also be implemented by one or more specially designed hardware and firmware components, such as a field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs) and the like in other embodiments. In an embodiment, the processor **200** may be, respectively, a special purpose processor which may be implemented via dedicated logic circuitry of an ASIC, an FPGA, or the like in order to enhance the processing speed of the operations discussed herein. The memory **204** also stores a repository **220** storing rules and data for the dimensioning operation.

[0024] The device **104** also includes a communications interface **224** enabling the device **104** to exchange data with other computing devices such as the server **124**. The communications interface **224** is interconnected with the processor **200** and includes suitable hardware (e.g., transmitters, receivers, network interface controllers and the like) allowing the device **104** to communicate with other computing devices-such as the server **124**. The specific components of the communications interface **224** are selected based on the type of network or other links that the device **104** is to communicate over.

[0025] The device **104** may further include one or more input and/or output devices **228**. The input devices may include one or more buttons, keypads, touch-sensitive display screens or the like for receiving input from a user. The output devices may further include one or more display screens, sound generators, vibrators, or the like for providing output or feedback to a user.

[0026] Turning now to FIG. **3**, the functionality implemented by the device **104** will be discussed in greater detail. FIG. **3** illustrates a method **300** of dimensioning a target object. The method **300** will be discussed in conjunction with its performance in the system **100**, and particularly by the device **104**, via execution of the application **208**. In particular, the method **300** will be described with reference to the components of FIGS. **1** and **2**. In other examples, some or all of the method **300** may be performed by other suitable devices or systems, such as the server **124**.

[0027] The method **300** is initiated at block **305**, where the device **104** obtains preliminary dimensioning data, and in particular, preliminary depth data from the field of view **116** of the depth sensor **112**. In particular, the field of view **116** may include the target object **108** to be dimensioned. For example, the device **104** may control the depth sensor **112** to capture the preliminary depth data from its field of view in response to a trigger from a user or in response to detecting a dimensioning condition. For example, the device **104** may process the depth data in the field of view and identify the dimensioning condition when an object **108**, or in particular, a cuboidal object **108**, is detected. In some examples, in addition to preliminary depth data, the device **104** may obtain additional preliminary dimensioning data, such as image data (e.g., optical image data, infrared data, or the like), other environmental data detected by additional sensors during the dimensioning operation, or the like.

[0028] In some examples, at block **305**, prior to obtaining the preliminary depth data, the device **104** may output, at the output device **228**, instructions for obtaining the preliminary depth data. For example, the device **104** may display text instructions and/or visual guidelines or produce an audio signal to prompt the user to orient the device **104** towards the target object **108**, such that the target object **108** is in the field of view **116** of the depth sensor **112**.

[0029] In particular, to facilitate the dimensioning operation described herein, the device **104** may display the field of view (e.g., as captured by an image sensor of the device **104**) with a region indicator of a predefined target region within that field of view **116**. The predefined target region represents a portion of the field of view **116** from which a subset of the preliminary depth data is to be extracted and validated for the dimensioning operation, as will be described herein. The predefined target region may be about centered in the field of view **116**, and may be substantially circular, rectangular, or have another suitable shape. The region indicator may therefore outline a boundary of the predefined target region to allow the user to visualize the predefined target region within the field of view **116** and align the target object **108** within the field of view **116** based on the region indicator.

[0030] At block **310**, the device **104** is configured to obtain a subset of the preliminary depth data from the predefined target region of the field of view **116**. For example, the device **104** may extract the depth data from pixels in the field of view **116** corresponding to the predefined target region.

[0031] At block **315**, the device **104** is configured to select a dimensioning function for the target object **108**. For example, the device **104** may select the dimensioning function based on the subset of preliminary depth data obtained at block **310**. In particular, the device **104** may be configured to identify or detect a geometrical element or feature (e.g., a vertex, an edge, a face, or the like) present in the subset of preliminary depth data an select the dimensioning function based on the detected geometrical element. For example, the device **104** may use one or more plane fitting functions, RGB (red-green-blue) and/or infrared-assisted boundary detection, and the like. The device **104** may further apply hand detection, point elimination, artificial depth hole filling functions to remove a hand of the user. Further, in some examples, the device **104** may apply detection functions to the preliminary dimensioning data from the entire field of view **116**, while applying the selection and validation criteria to the subset of preliminary depth data from the predefined target region.

[0032] In other examples, the device **104** may select the dimensioning function based on a selection input from the user of the device **104**. The dimensioning functions may include a perpendicular-face approach, a two-sided perspective approach, a three-sided perspective approach, or similar. That is, the dimensioning function may vary by a target orientation of the object **108** to be dimensioned. Accordingly, in addition to selecting a dimensioning function for the target object **108**, the device **104** may additionally identify a target orientation for the target object **108** to allow the target object **108** to be successfully dimensioned using the selected dimensioning function.

[0033] At block **320**, the device **104** is configured to validate that the target object **108** is in the target orientation using the subset of the preliminary depth data obtained at block **310**. For example, the device **104** may analyze the subset of the preliminary depth data to identify geometrical elements or features of the target object **108** and verify that appropriate geometrical elements are present according to the target orientation.

[0034] For example, referring to FIG. **4**, a flowchart of an example method **400** of selecting a dimensioning function and validating the target orientation for the dimensioning function is depicted.

[0035] At block **405**, the device **104** determines whether a vertex is detected in the subset of the preliminary depth data. For example, the device **104** may use any suitable functions configured to analyze depth data and identify geometric features, including vertices and/or corners of objects.

[0036] If the determination at block **405** is affirmative, the device **104** proceeds to block **415-1**. At block **415-1**, the device **104** is configured to select the three-sided perspective approach as the dimensioning function. That is, having identified a vertex in the predefined target region, the device **104** may assume that a user has or is orienting the target object **108** to allow the device **104** to have a view of the target object **108** in which three of the faces are visible in the field of view **116**.

[0037] Accordingly, at block **420**, the device **104** is configured to validate the target orientation for the selected three-sided perspective approach. In particular, for the three-sided perspective approach, the device **104** may verify that in addition to a vertex, three distinct faces are detectable in the predefined target region. That is, the target orientation is a perspective view including three sides of the object. The device **104** may further validate that the three distinct faces are oriented substantially perpendicular to one another. If the device **104** validates that the target object **108** achieves the target orientation for the three-sided perspective approach, the device **104** may proceed to block **325** of the method **300**.

[0038] Thus, for example referring to FIG. **5**A, the dimensioning device **104** is depicted with a display **500** configured to display a representation of the field of view **116**. The display **500** may further be configured to display a region indicator **504** representing the predefined target region **508**. Accordingly, in operation, at block **405-1**, the dimensioning device **104** may detect a vertex **512** of an object **516-1** in the preliminary depth data corresponding to the predefined target region **508**. At block **420**, the dimensioning device **104** may further verify that three distinct faces **520-1**, **520-2**, and **520-3** are detected in the preliminary depth data corresponding to the predefined target region **508**. The device **104** may then validate that the target orientation for the three-sided perspective approach as the dimensioning function has been achieved, and may proceed to block **325** of the method **300**.

[0039] Returning to FIG. **4**, if the determination at block **405** is negative, the device **104** proceeds to block **410**. At block **410**, the device **104** determines whether an edge is detected in the subset of preliminary depth data. Further, the device **104** may determine whether the edge extends across the predefined target region, from one point on the perimeter of the predefined target region to another point on the perimeter of the predefined target region (e.g., the edge represents a chord or segment extending across the predefined target region).

[0040] If the determination at block **410** is affirmative, the device **104** proceeds to block **415-2**. At block **415-2**, the device **104** is configured to select the two-sided perspective approach as the dimensioning function. That is, having identified an edge in the predefined target region, the device **104** may assume that a user has or is orienting the target object **108** to allow the device **104** to have a view of the target object **108** in which two of the faces are visible in the field of view **116**. In particular, in the target orientation, the object **108** may be oriented in such a way that only two of the faces are visible in the field of view **116**.

[0041] Accordingly, at block **420**, the device **104** is configured to validate the target orientation for the selected two-sided perspective approach. In particular, for the two-sided perspective approach, the device **104** may verify that in addition to an edge, two distinct faces are detectable in the predefined target region. That is, the target orientation is a perspective view including two sides of the object. The device **104** may further validate that the two distinct faces are oriented substantially perpendicular to one another. If the device **104** validates that the target object **108** achieves the target orientation for the two-sided perspective approach, the device **104** may proceed to block **325** of the method **300**.

[0042] For example, referring to FIG. **5**B, the dimensioning device **104** is depicted dimensioning an object **516-2**. In this example, at block **405-1**, the dimensioning device **104** may not detect a vertex, and at block **405-2**, the dimensioning device **104** may detect an edge **524** of the object **516-2** in the preliminary depth data corresponding to the predefined target region **508**. However, based on the orientation of the object **516-2**, only one face of the object **516-2** may be detected in the preliminary depth data corresponding to the predefined target region **508**. That is, while the edge **524** may extend on one side to a face of the object **516-2**, the edge **524** may represent a boundary between the face and an environment of the object **516-2** visible in a background of the field of view **116**. Accordingly, the validation at block **420** may fail.

[0043] Returning again to FIG. **4**, if the determination at block **410** is negative, the device **104** may proceed to block **415-3**. At block **415-3**, the device **104** is configured to select the perpendicular-face approach. That is, the device **104** may assume that a user has or is orienting the target object **108** to dimension one face of the target object **108** at a time.

[0044] Accordingly, at block **420**, the device **104** may validate the target orientation for the selected perpendicular-face approach. In particular, for the perpendicular-face approach, the device **104** may validate that the single face or surface is detected in the predefined target region, and in some examples, whether an average depth of the single face or surface is within a threshold distance of the device **104**, for example to verify that the target object **108** is sufficiently close to dimension. In some examples, the device **104** may additionally validate an orientation angle of the target object **108** as described below in conjunction with the perpendicular-face dimensioning operation. For example, block **420** may be integrated into the performance of the method **600** as described below. If the device **104** validates that the target object **108** achieves the target orientation for the perpendicular-face approach, the device **104** may proceed to block **325** of the method **300**.

[0045] If the validation block **420** fails, then the device **104** may identify an error condition at block **425**. In some examples, the device **104** may generate an error condition alert, such as a visual indication (e.g., at an indicator light, a display screen or the like), an audio indication, a notification, or similar. Alternately, the device **104** may simply wait for the object to be repositioned to a valid orientation for the selected dimensioning function or for another dimensioning function.

[0046] Returning to FIG. **3**, in some examples, in addition to validating the target orientation of the object **108** at block **320**, the device **104** may perform other validations. For example, the device **104** may validate that the target object **108** occupies a threshold proportion of the field of view **116**. The device **104** may further validate that an entirety of the target object **108** is visible in the field of view. For example, the device **104** may validate that a boundary of the target object **108** does not coincide with the boundary of the frame of view **116**.

[0047] At block **325**, the device **104** is configured to apply the selected dimensioning function to dimension the target object. In some examples, the device **104** may obtain further dimensioning data, for example by controlling the depth sensor **112** and/or any other additional sensors to capture data from the field of view **116**. In particular, since the method **300** may particularly be applied to hand-held and/or small objects **108**, the device **104** may first segment the captured data to identify the object to be dimensioned. The device **104** may assume that the object **108** remains substantially covering the predefined target region, and hence may discard the dimensioning data not identified as being part of the object **108** originating in the predefined target region. That is, the dimensioning function may be configured to be applied using the segment representing the target object only, rather than being based on relative support surfaces such as a floor, table, or the like. As a result, the device **104** may discard the segments of the depth data having depth values only outside the predefined target region, which may represent, for example, the environment of the object including any detected support surfaces, other objects, or the like. Thus, when the object **108** is handheld, the device **104** may be enabled to detect and dimensions objects **108** which are substantially flat, such as letters or envelopes, since the handheld nature of the object **108** allows the boundaries of the substantially flat objects to be distinguished from other substantially planar surfaces, such as the floor or other support surfaces.

[0048] The device **104** may further apply one or more hand detection and point elimination functions, as well as hole filling functions to eliminate depth data from a user's hand holding the object **108**.

[0049] The device **104** may then apply the selected dimensioning function to determine dimensions of the object **108**. For the three-sided and two-sided perspective approaches, all three dimensions (i.e., length, width and height) of the cuboidal object **108** are captured in the field of view when the object **108** is in the target orientation, and hence the device **104** may simply use the obtained depth data to determine the dimensions of the object **108**.

[0050] In the perpendicular-face approach, the device **104** may be configured to obtain first depth data of a first surface when the first surface is perpendicular to the principal axis and second depth data of a second surface when the second surface is perpendicular to the principal axis. That is, two dimensions may be detectable in the field of view **116** in the first step when the first surface is perpendicular to the principal axis and a different set of two dimensions may be detectable in the field of view **116** at the second step when the second surface is perpendicular to the principal axis. Accordingly, the device **104** may prompt the user to move the target object **108** to allow a different surface to be visible. Further, surface may be easier to dimension when the surface is substantially perpendicular to the principal axis **120** of the depth sensor **112**, and hence the device **104** may facilitate alignment of the target object **108** to be substantially perpendicular to the principal axis **120** of the depth sensor **112**. For example, referring to FIG. **6**, an example method **600** of applying a perpendicular-face approach dimensioning function is depicted.

[0051] At block **605**, the device **104** may obtain, segment and filter dimensioning data, and in particular, depth data, from the field of view **116**. In particular, the device **104** may segment and filter the depth data to retain the surface from the predefined target region. The device **104** may similarly apply hand detection, point elimination and hole filling functions to remove a user's hand from the dimensioning data.

[0052] At block **610**, the device **104** determines an angle of the surface relative to the principal axis **120** of the depth sensor **112**. For example, the device **104** may apply plane fitting and determine a surface normal of the surface, and then compute the relative angle of the surface normal to the principal axis **120**. In other examples, the device **104** may identify the four vertices of the surface (i.e., assuming a cuboidal object **108**) and use the respective depths to determine the angle of the surface. In other examples, the device **104** may employ one or more edge detection, plane fitting or other suitable functions to facilitate the computation of the angle of the surface relative to the principal axis **120** of the depth sensor **112**.

[0053] At block **615**, the device **104** is configured to display an alignment indicator based on the angle of the surface computed at block **610**. In particular, the alignment indicator may be displayed on a display of the device **104**. The alignment indicator may be positioned relative to the region indicator according to the angle of the surface, such that when the surface is perpendicular to the principal axis **120** of the depth sensor **112**, the alignment indicator is centralized in the region indicator for the predefined target region. When the surface normal of the detected surface has an angular offset from the principal axis **120**, the alignment indicator may be similarly offset from the region indicator in the direction of the angle offset. Thus, for example, the alignment indicator may have a shape complementary to that of the region indicator.

[0054] For example, referring to FIG. **7**A, the device **104** displays the field of view **116** with an object **700** having a surface **704** oriented at an angle to the principal axis **120** of the depth sensor **112**. Upon detecting in the predefined target region, as indicated by a region indicator **708**, that the surface **704** is detected, the device **104** may compute the angle of the surface **704** and additionally display an alignment indicator **712**, which is misaligned with the region indicator **708** to indicate the misalignment of the surface **704** relative to the principal axis **120**. For example, the alignment indicator **712** may be skewed towards a portion of the surface **704** which is further away from the depth sensor **112**.

[0055] In FIG. **7**B, a user may adjust the position of the object **700** to be substantially perpendicular to the principal axis **120**, and accordingly the alignment indicator **712** is aligned with the region indicator **708**.

[0056] Returning to FIG. **6**, at block **620**, the device **104** determines whether the alignment indicator is aligned with the region indicator. In some examples, the determination at block **620** may be performed automatically by the device **104**, while in other examples, the determination at block **620** may be performed with input from the user of the device **104**. If the determination at block **620** is negative, then the device **104** may continue to periodically calculate the angle of the surface, update the display of the alignment indicator and wait until the alignment indicator is aligned with the region indicator.
[0057] If the determination at block **620** is affirmative, the device **104** may proceed to block **625**. At block **625**, the device **104** may be configured to obtain dimensioning data, for example including depth data, image data, or the like. In particular, upon detecting alignment of the alignment indicator with the region indicator, the device **104** may control the depth sensor **112** (and any further sensors of the device **104**) to capture dimensioning data while the target object **108** is oriented perpendicular to the depth sensor **112**. The device **104** may be controlled to capture the dimensioning data automatically or in response to an input from the user.
[0058] At block **630**, the device **104** determines whether sufficient dimensioning data from the perpendicular faces has been obtained to allow the device **104** to complete the dimensioning operation. For example, if dimensioning data has been obtained from two perpendicular faces, the device **104** may determine that the dimensioning operation may be completed. In other examples, the device **104** may receive an input from a user that only one perpendicular face is to be dimensioned, for example if the object **108** is a substantially flat object, such as a letter or envelope or similar.
[0059] If the determination at block **630** is negative, that is, only one perpendicular face has been dimensioned, and the device **104** has received no indication from the user of completion, then the device **104** proceeds to block **635**. At block **635**, the device **104** may prompt the user to align the second face perpendicular to the principal axis. For example, the device **104** may display a prompt, provide an audio indication, or similar. In some examples, the device **104** may wait for a confirmation from the user that the second face is ready for dimensioning before returning to block **610** to facilitate aligning the second face perpendicular to the principal axis **120**.
[0060] If the determination at block **630** is affirmative, then the device **104** proceeds to block **640**. At block **640**, the device **104** is configured to determine dimensions for the object. In particular, each face presented perpendicular to the principal axis **120** allows the object to be dimensioned along two axes. When two faces are presented, the device **104** may expect one axis from each face to be common, and hence may identify the dimensions with the least difference as being along the same axis of the object **108**. In some examples, when the closest dimensions are outside a threshold similarity, the device **104** may identify an error condition.
[0061] Upon determining the dimensions at block **640** or block **325**, the device **104** may store the dimensions of the object in the repository **220**, present the dimensions at the display of the dimensioning device **104**, send the dimensions to the server **124** for further storage or processing, or similar.
[0062] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.
[0063] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.
[0064] Moreover in this document, relational terms such as first and second, top and bottom, and

the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has”, “having,” “includes”, “including,” “contains”, “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0065] It will be appreciated that some embodiments may be comprised of one or more specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used.

[0066] Moreover, an embodiment can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g., comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

[0067] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A dimensioning device comprising: a depth sensor configured to obtain depth data within a field of view; a processor interconnected with the depth sensor, the processor configured to: obtain depth data within the field of view including an object to be dimensioned; obtain a subset of the depth data from a predefined target region within the field of view; select a dimensioning function for the object, the dimensioning function having a target orientation of the object; validate the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, apply the dimensioning function to obtain dimensions of the object.

2. The dimensioning device of claim 1, wherein the processor is configured to select the dimensioning function for the object by: detecting an element of the object in the target region; and selecting the dimensioning function based on the detected element of the object.

3. The dimensioning device of claim 1, wherein the processor is further configured to display a region indicator representing the target region at a display of the dimensioning device.

4. The dimensioning device of claim 1, wherein the dimensioning function comprises a perpendicular-face approach, and wherein the target orientation includes an orientation of a surface perpendicular to a principal axis of the depth sensor.

5. The dimensioning device of claim 4, wherein to apply the dimensioning function, the processor is configured to: obtain first depth data of a first surface of the object when the first surface is perpendicular to the principal axis; obtain second depth data of a second surface of the object when the second surface is perpendicular to the principal axis; and determine the dimensions of the object based on the first depth data and the second depth data.

6. The dimensioning device of claim 4, wherein the processor is configured to: determine an angle of the surface relative to the principal axis of the depth sensor; and display an alignment indicator at a display of the dimensioning device based on the determined angle.

7. The dimensioning device of claim 6, wherein the alignment indicator is offset from an indicator of the target region based on the determined angle.

8. The dimensioning device of claim 1, wherein the dimensioning function comprises a two-sided perspective approach, and wherein the target orientation of the object results in a perspective view including two sides of the object.

9. The dimensioning device of claim 1, wherein the dimensioning function comprises a three-sided perspective approach, and wherein the target orientation of the object results in a perspective view including three sides of the object.

10. A method comprising: obtaining depth data within a field of view including an object to be dimensioned; obtaining a subset of the depth data from a predefined target region within the field of view; selecting a dimensioning function for the object, the dimensioning function having a target orientation of the object; validating the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, applying the dimensioning function to obtain dimensions of the object.

11. The method of claim 10, comprising selecting the dimensioning function for the object by: detecting an element of the object in the target region; and selecting the dimensioning function based on the detected element of the object.

12. The method of claim 10, further comprising displaying a region indicator representing the target region at a dimensioning device configured to dimension the object.

13. The method of claim 10, wherein the dimensioning function comprises a perpendicular-face approach, and wherein the target orientation includes an orientation of a surface perpendicular to a principal axis of a depth sensor of a dimensioning device configured to dimension the object.

14. The method of claim 13, wherein applying the dimensioning function comprises: obtaining first depth data of a first surface of the object when the first surface is perpendicular to the principal axis; obtaining second depth data of a second surface of the object when the second surface is perpendicular to the principal axis; and determining the dimensions of the object based on the first depth data and the second depth data.

15. The method of claim 13, further comprising: determining an angle of the surface relative to the principal axis of the depth sensor; and displaying an alignment indicator at the dimensioning device based on the determined angle.

16. The method of claim 15, further comprising offsetting the alignment indicator from an indicator of the target region based on the determined angle.

17. The method of claim 10, wherein the dimensioning function comprises a two-sided perspective approach, and wherein the target orientation of the object results in a perspective view including two sides of the object.

18. The method of claim 10, wherein the dimensioning function comprises a three-sided perspective approach, and wherein the target orientation of the object results in a perspective view including three sides of the object.

19. A non-transitory computer-readable storage medium storing instructions thereon, which when executed configure a processor to: obtain depth data within a field of view including an object to be dimensioned; obtain a subset of the depth data from a predefined target region within the field of view; select a dimensioning function for the object, the dimensioning function having a target orientation of the object; validate the target orientation of the object based on the subset of the depth data in the predefined target region; and when the target orientation is validated, apply the dimensioning function to obtain dimensions of the object.

20. The non-transitory computer-readable storage medium of claim 19, wherein the instructions further configure the processor to: segment the depth data and discard segments of the depth data outside the predefined target region; and apply the dimensioning function a segment of the depth data originating in the predefined target region.